

**1. Prices are normally distributed**—Options are priced based on the assumption of a normal distribution, which effectively implies that future prices near the current level are most probable and that probabilities drop steeply for prices further removed from current levels. In some instances, however, large price moves are much more likely than implied by the normal distribution. Consider, for example, Mai's comments regarding Capital One: "The odds that the stock would still be near \$30 in two years seemed vanishingly small. Either the company would be vindicated or it would go under." In such circumstances, options priced in line with option-pricing models will be severely mispriced; specifically, out-of-the-money options will be priced far too cheaply. These mispricings can create profit opportunities. If the probability of a large price move is much greater than normal, as was the case with Capital One, then out-of-the-money options can provide a significant probability of gain, while being priced consistent with long-shot probabilities.

Another implication of the normal distribution assumption is that it is always equally likely for prices to go up X percent as down X percent. Although this assumption may often be a reasonable approximation, there are times when it is far more likely for a market to go up by a given percentage than down by the same amount or vice versa. A good example of such an asymmetric price outlook was the Korean stock market when Mai implemented his long position. At the time, the market was not only severely undervalued relative to other Asian markets, but prices were so low that many companies had market capitalizations that were less than their net cash balances—that is, the companies were effectively being valued at less than zero. In such circumstances, the odds of a large price advance are significantly greater than the odds of an equivalent price decline.

There is an important corollary here: The contention by many academics that the market price is always right is inconsistent with the empirical evidence. There are times when the market price is clearly wrong, and these situations provide major trade opportunities.